

Forces shaping genetic diversity

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An introduction to population genetics through coalescent theory

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What is Population Genetics?

“A mathematical / statistical description of evolution”

The goal of population genetics is to develop mathematical and statistical models of evolution and to use them to

- to learn about the **evolutionary history** of a species or population
- to study **evolutionary processes** in nature.

What is evolution?

Any **change** in the frequency of a genetic trait between generations.

Note: Evolution does not require adaptation!

Important definitions

Locus (plural loci)

A locus is the specific position or consecutive positions on a chromosome. When we talk about a locus, we may thus mean a gene or a specific SNV.

Allele

An allele is one of a number of alternative forms of the same locus.

At a specific SNV, the bases A and C are different alleles.

Haploid - Diploid

The ploidy indicates the number of copies of a locus an individual carries.

Haploid refers to a single copy, diploid to two copies.

Humans, for instance, are diploid at most loci (but not on sex chromosomes).

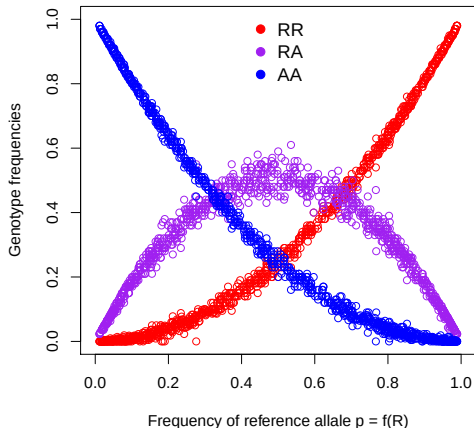
Genotype

The genotype is the collection of alleles of an individual at one locus.

Homozygous genotypes: two identical alleles at a diploid locus (e.g. two A).

Heterozygous genotypes: two different alleles (e.g. A and C).

Relationship between Genotype and Allele Frequencies



The genotype frequencies as a function of the allele frequency $p = f_R$ for each locus.

Questions:

Why do we see these *striking* regularities?

Can we build a simple model that explains these observations?

Building a theoretical model for genotype frequencies

Basic assumptions

- Individuals result from the fusion of two gametes (e.g. sperm and egg).
- Each gamete contributes one allele.
- At a heterozygous locus, both allele are transmitted with equal probability.
- Mating is random (i.e. independent of the allele).

Expected gamete frequencies

Given genotype frequencies f_{RR} , f_{RA} and f_{AA} , the probability that a random gamete carries the reference (R) allele is $f_{RR} + \frac{1}{2}f_{RA} = p$.

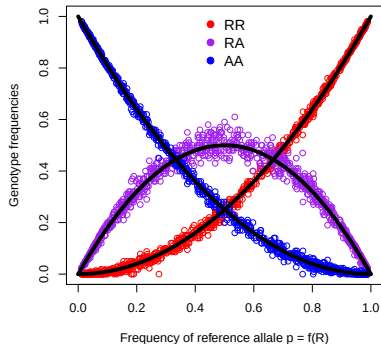
Expected genotypes after random mating

$$f'_{RR} = p^2, f'_{RA} = 2p(1 - p) \text{ and } f'_{AA} = (1 - p)^2$$

Expected allele frequency in the next generation

$$p' = f'_{RR} + \frac{1}{2}f'_{RA} = p^2 + \frac{1}{2}(2pq) = p$$

Model versus Data



Implications of this model

- 1) Allele frequencies **don't change!**
 $p' = p$ and $q' = q$.
- 2) The expected genotype frequencies are p^2 , $2pq$ and q^2 .
- 3) They are reached after **one generation** of random mating!

Q: Do our calculations check out?

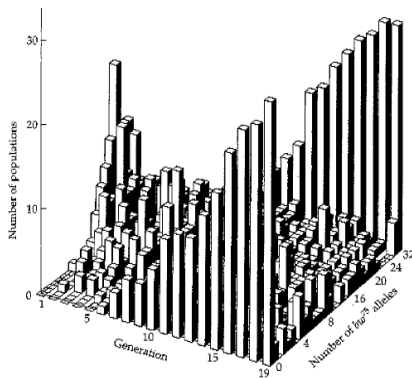
Loci showing these frequencies are said to be in **Hardy-Weinberg equilibrium (HWE)**.

Allele frequencies DO change over time!

Experiment with *D. melanogaster*

In a classic experiment, 107 populations of 16 individuals of *D. melanogaster* were kept in vials for 19 generations.

After reproduction, 8 males and 8 females were randomly picked to form the next generation.



HWE is an expectation

The expectation of a random process

The expectation of a random process is the average outcome if we were to replicate the process infinitely many times.

Example: if $f_A = 0.5$, how many heterozygotes do we expect among 1 individual?

Answer: $2f_Af_a = 0.5$.

Hence we sometimes see 1, sometimes 0, but *on average* 0.5

HWE is an expectation

Under HWE we *expect* the allele frequency not to change, but in reality we will see random fluctuations.

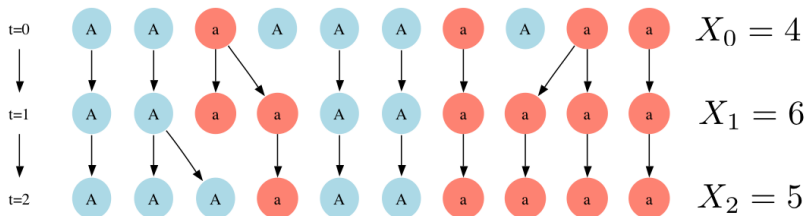
Definition: Genetic Drift

A change in allele frequency because alleles *by chance* produce more / less offspring in any given generation.

The Wright-Fisher (WF) model

Basic assumption: random picking of gametes

The WF-model assumes that N_t adults produce a large number of gametes, of which $N_t + 1$ are chosen randomly to constitute the next generation $t + 1$.



Definition Genetic Drift

Genetic drift: a change in allele frequency because alleles *by chance* produce more / less offspring in any given generation.

The Wright-Fisher (WF) model

Simulating the WF-model

Since the WF-model only assumes that gametes (and hence parents) are picked at random, it is super simple to simulate.

Transition probabilities

If i out of N_t adults carry allele a , and thus a proportion $p = \frac{i}{N}$ of all gametes will be of type a , then

$$P(X_{t+1} = j | X_t = i) = \binom{N}{j} p^j (1 - p)^{N-j}$$

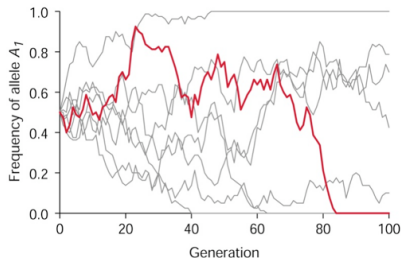
where $\binom{N}{j}$ is the number of ways to arrange j a alleles among N gametes.

Note: this distribution is known as the *binomial sampling distribution*.

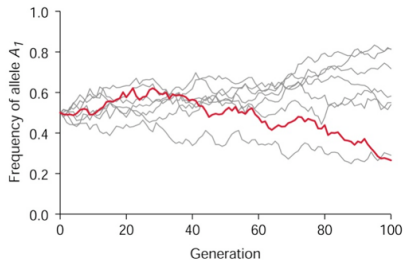
Simulating the WF-model

Examples of simulated trajectories

(b) Population size = 40



(c) Population size = 400



Note: the magnitude of the change is a function of the population size.

Note: under drift alone, every allele will eventually become fixed or lost!

Questions: Why is there still genetic diversity? And how does it arise?

Mutation rates

Modelling mutations

Mutations come in many form, but most can be modeled with two alleles.

How frequent do mutations occur?

Mutation rates for Single Nucleotide Variants (SNV) estimated from Icelandic pedigrees.

Type of mutation	<i>n</i>	Rate per base per generation
Transition at non-CpG	2,489	6.18×10^{-9}
Transition at CpG	855	1.12×10^{-7}
Transversion at non-CpG	1,516	3.76×10^{-9}
Transversion at CpG	73	9.59×10^{-9}
All	4,933	1.20×10^{-8}

Mutation rates are per generation per base. For non-CpG sites, the effective number of bases examined is taken as 2.583 billion, whereas for CpG sites the number is 48.8 million. These numbers take into account the variation of local coverage in sequencing (Supplementary Information).

Mutation rates are on the order of $\mu = 1.2 \cdot 10^{-8}$ per generation per base pair.

What does $\mu = 1.2 \cdot 10^{-8}$ imply?

- Mutations are very rare. Imagine doing a single spelling mistake when copying LOTR 26 times by hand!
- But since our genome is $3.2 \cdot 10^9$ base pairs long, we still expect $2 \cdot 3.2 \cdot 1.2 \cdot 10^9 \cdot 10^{-8} = 76.7$ new mutations per individual per generation!
- Since we are $8 \cdot 10^9$ individuals on this planet, the probability m that no single person has a mutation at a specific base pair is

$$m = (1 - 1.2 \cdot 10^{-8})^{2 \cdot 8 \cdot 10^9} = 4.1 \cdot 10^{-84},$$

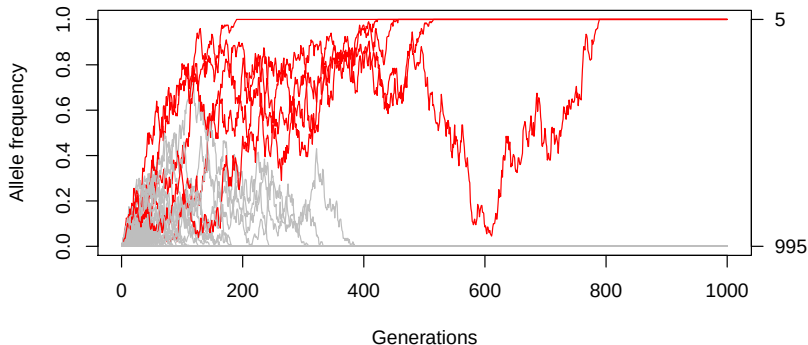
which is less likely that two people pick the the same atom in the universe at random!

Fate of a new mutation

Most new mutations will disappear again

Eventually, all mutations will either get fixed or be lost.

Most new mutations will be lost.



Question: what is the probability that a new mutation will fix?

Fate of a new mutation

Probability of fixation

- Each allele will eventually be fixed or get lost.
- This is also true if we define an allele as the copy of one individual.
- In the absence of selection, each such allele should have the same probability to become fixed. And if one fixes, all others will be lost.
- Hence, the probability that the allele of one individual will fix is $\frac{1}{2N}$ in a diploid population.
- If an allele is present in N_A copies, it's probability of fixation is $\frac{N_A}{2N} = f_A$

Under genetic drift, the probability of fixation is simply the allele frequency!

Rate of Substitution

A Substitution...

... occurs if an ancestral allele is replaced by a novel, derived allele.

Question: what is the rate of substitution?

- Each gene copy has a probability μ to be affected by a mutation. Hence, the expected number of mutations entering the population per generation is $2N\mu$.
- Each of those new mutations has a probability of $\frac{1}{2N}$ to become fixed.
- The expected number of mutations that enter the population each generation and will eventually fixed (the number of substitutions S) is thus

$$E(S) = 2N\mu \cdot \frac{1}{2N} = \mu$$

Under genetic drift, the rate of substitution is **equal to mutation rate** and **independent of the population size!** → Molecular clock.

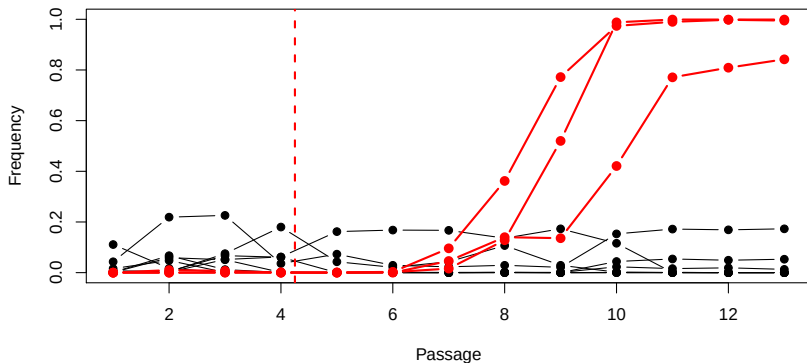
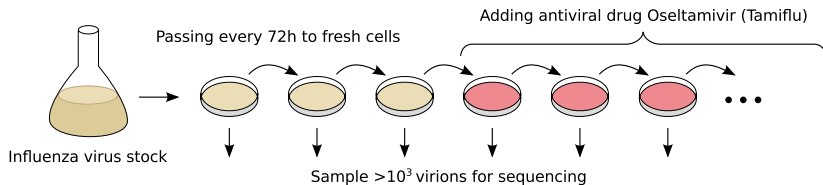
We have seen that:

- Over time, genetic drift is reducing genetic diversity.
- The reason we still observe genetic diversity are new mutations.

Question: How much genetic diversity does a population have?

This is best understood using coalescent theory → look forward to the afternoon!

Evolution of Drug Resistance in Influenza



Simple model of selection

Individual Fitness

The reproductive contribution of an individual to the next generation. Selection results as differences in fitness between individuals.

Selection acts on the Phenotype

Selection can only act on phenotypic differences between individuals. A change in the phenotype over time only occurs if the phenotype is *heritable*.

Genotypic Fitness λ

To study the effect of selection on allele frequencies, we define the *genotypic fitness* λ as the average fitness of all individuals who have the same genotype.

Simple model of selection

A simple model for haploid selection

- N haploid individuals, of which N_A carry the A and N_a the a allele.
- Each individual has the chance to survive and have offspring.
- On average, A -bearing individuals have λ_A and a -bearing λ_a descendants.

Allele frequency change due to selection

$$f_A(0) = \frac{N_A}{N_A + N_a}$$

$$f_A(1) = \frac{\lambda_A N_A}{\lambda_A N_A + \lambda_a N_a} = \frac{\lambda_A f_A(0)}{\lambda_A f_A(0) + \lambda_a f_a(0)} = \frac{f_A(0)}{f_A(0) + \frac{\lambda_a}{\lambda_A} f_a(0)}$$

Note: allele frequency changes depend on $\frac{\lambda_a}{\lambda_A}$, not the absolute values!

Simple model of selection

Selection coefficient s and relative fitness ω

Relative fitnesses: $\omega_A = 1$ and $\omega_a = 1 - s$, where s is the selection coefficient.

Note that $\frac{\lambda_a}{\lambda_A} = \frac{\omega_a}{\omega_A} = 1 - s$, and hence $f_A(1) = \frac{f_A(0)}{f_A(0) + (1 - s)f_a(0)}$

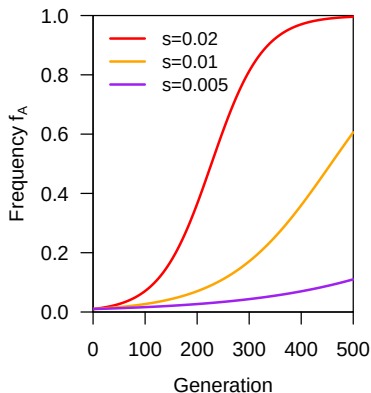
Change over multiple generations

After t time steps, we get:

$$f_A(t) = \frac{f_A(0)}{f_A(0) + (1 - s)^t f_a(0)}$$

Note: the change in frequency is strongest when $f_A \approx \frac{1}{2}$.

Rule of thumb: it takes about $\frac{1}{s}$ generations for selection to change the allele frequency substantially.

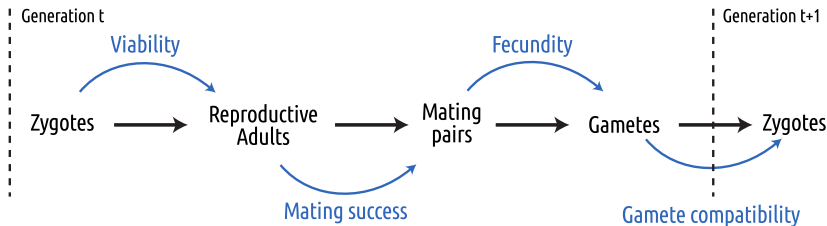


Selection in diploids

Selection in diploids

A model of selection for diploid organisms is a combination of Hardy-Weinberg model of sexual reproduction and genotypic selection.

Selection might act at any point in the life cycle of an organism.



Viability selection

Selection resulting from differences in individuals in their abilities to survive and reproduce (“Struggle for existence” in Darwin’s terms).

Note: Viability selection is the easiest to work with, but the principles are the same for most kinds of selection.

Viability selection: a simple model

Step 1: Random mating

If the allele frequencies among the parents are $f_A(t)$ and $f_a(t) = 1 - f_A(t)$, the genotype frequencies among the zygotes are $f_A(t)^2$, $2f_A(t)f_a(t)$ and $f_a(t)^2$.

Step 2: Selection

The probabilities of survival (viabilities) for the three genotypes are denoted by v_{AA} , v_{Aa} and v_{aa} . The allele frequencies in the next generation are then:

$$f_A(t+1) = \frac{v_{AA}f_A(t)^2 + v_{Aa}f_A(t)f_a(t)}{v_{AA}f_A(t)^2 + 2v_{Aa}f_A(t)f_a(t) + v_{aa}f_a(t)^2}$$

Note: again, only the relative viabilities matter!

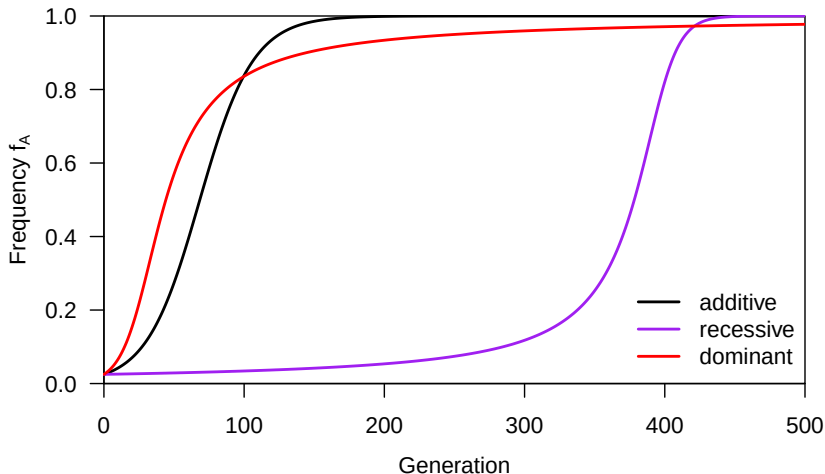
Special cases	ω_{AA}	ω_{Aa}	ω_{aa}
Genic selection	1	$1 - s$	$(1 - s)^2$
Additive selection	1	$1 - s$	$1 - 2s \approx (1 - s)^2$
Dominant selection	1	1	$1 - s$
Recessive selection	1	$1 - s$	$1 - s$

Directional selection

Directional selection

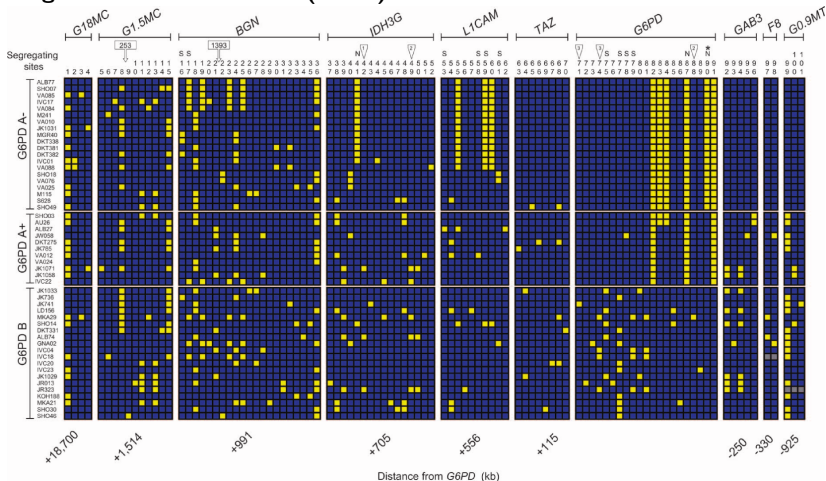
Selection that is favoring one allele over another: $v_{aa} \leq v_{Aa} \leq v_{AA}$.

Example with $s_{aa} = 0.1$:



Loci are not independent but linked

Single Nucleotide Variants (SNVs) in and around G6PD



Note: SNVs are not independent, they are in linkage disequilibrium (LD)!

Measuring Linkage Disequilibrium

Haplotype

A specific combination of alleles on the same chromosome. In case of two loci with alleles A/a and B/b , the possible haplotypes are AB , Ab , aB and ab .

Expected haplotype frequencies under independence

When two loci are segregating independently, we expect a fraction $\mathbb{E}[f_{AB}] = f_A f_B$ of all haplotypes to be AB .

Coefficient of LD D

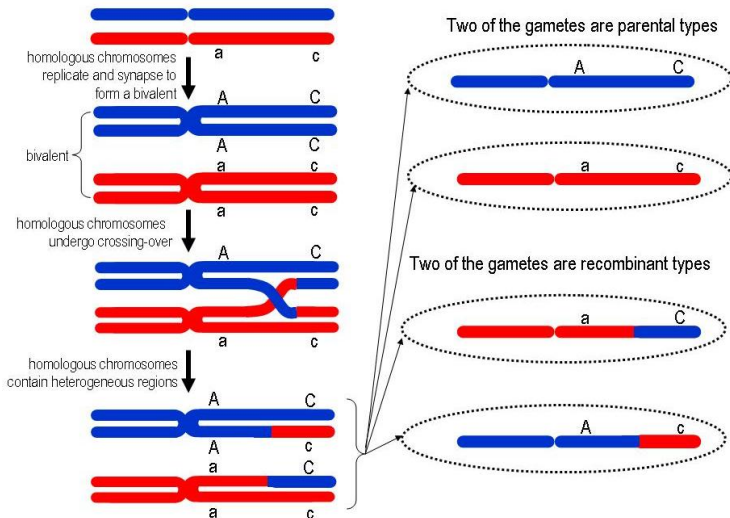
$D = |f_{AB} - f_A f_B| = |f_{Ab} - f_A f_b| = |f_{aB} - f_a f_B| = |f_{ab} - f_a f_b|$ measures the difference between haplotype frequencies and their expectation under independence.

If $D = 0$, the alleles A and B are said to be in linkage equilibrium.

D is maximal if at least one haplotype is missing from the sample.

Note: this implies that in case of a new mutation, D is maximized!

Recombination breaks haplotypes apart



Recombination breaks haplotypes apart

Recombination rate c

The recombination rate c is the fraction of recombinant gametes produced by a double heterozygote (e.g. $AaBb$).

If two loci are independent, a double heterozygote (e.g. $AaBb$) produces these four gametes in equal proportions (according to Mendel): AB , Ab , aB and ab .

Hence: $0 \leq c \leq \frac{1}{2}$, where $c = \frac{1}{2}$ indicates independence.

Evolution of D

It is easy to show that $\mathbb{E}[D(t)] = (1 - c)^t D(0)$.

What this means:

In humans, $c \approx 0.01$ for two loci 1 Mb (10^6 base pairs) apart.

For two loci 1Mb apart, it thus takes about 300 generation to reduce D to 10% of its original value.

For two loci in neighboring genes ($\approx \frac{1}{7}$ Mb apart), it takes > 1600 generations or $> 40,000$ years!

Genetic hitchhiking

Selective sweep

If an advantageous mutation is driven to fixation.

Partial sweep

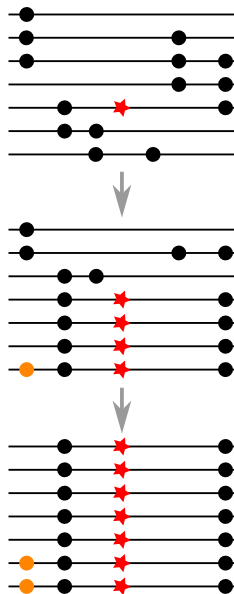
If an advantageous mutation has increased substantially in frequency but is not yet fixed.

Genetic Hitchhiking

During a selective sweep, linked neutral mutations get a free ride to higher frequency.

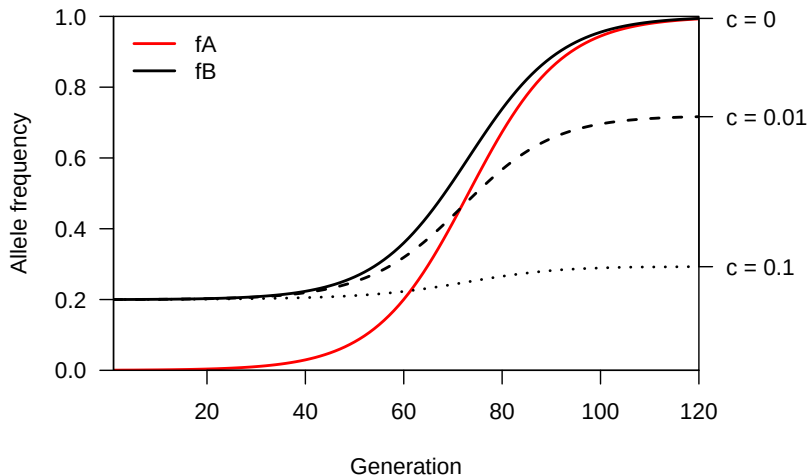
As a result, most of the genetic diversity in the region is lost.

Some neutral mutations may recombine onto a haplotype with the advantageous alleles and hence escape being lost.



Genetic hitchhiking

Numerical iteration of a new mutation for $N = 1000$, $fB(0) = 0.2$ and $s = 0.1$.



Note: These are expected trajectories in the absence of drift!

Selection versus Drift

Recap Selection

- Selection may lead to more rapid changes in allele frequencies than expected under drift alone.
- Allele frequency changes due to selection are driven by the relative fitness of the different genotypes.
- While dominant mutations increase the most quickly in frequency, they are the slowest to fix!
- Neutral mutations linked to selected mutations may hitchhike.

Confounding factor: genetic drift

This simple model of selection ignores *genetic drift*, but allele frequencies also change due to drift.

Question: how to differentiate drift from selection?

We need a statistical test!

In order to decide if a locus under selection, we need to show that a model with selection is more likely than a model with drift alone.

Fixation probabilities

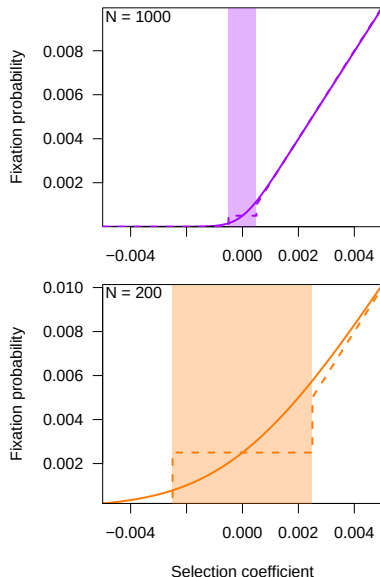
Fixation probability

The probability of fixation of a new mutation is given by

$$u(s, N) = \frac{1 - e^{-2s}}{1 - e^{-4Ns}}$$

Three domains

- 1) If $2Ns > 1$, the mutation is considered *strongly advantageous* and $u(s, N) \approx 2s$.
- 2) If $2Ns < -1$, the mutation is considered *strongly deleterious* and $u(s, N) \approx 0$.
- 3) If $-1 < 2Ns < 1$, the mutation is considered *nearly neutral* and $u(s, N) \approx \frac{1}{2N}$.



Effect of genetic drift on selected mutations

- Most advantageous mutations will be lost (unless $|s| > 0.25$).
- Some deleterious mutations will fix by chance.
- The population size has a strong effect on *nearly neutral* mutations ($-1 < 2Ns < 1$), but not on *strongly* selected mutations.
- **BUT:** the nearly neutral range depends on N !
⇒ Selection is much more efficient in large populations.

Summary

- Genetic Drift: Allele frequency changes due to random sampling in finite populations.
- Genetic drift reduces genetic diversity. Mutations regenerate genetic diversity.
- The rate of substitutions is the mutation rate and independent of population size.
- Selection may favor one allele over another, skewing fixation probabilities.
- Alleles are experiencing drift and selection simultaneously
→ selection is more efficient in large populations.